

Corrected Pseudo-Label Self-Training for Monocular Pallet Pose Estimation under an Explicit Real-Supervision Budget

Technical manuscript for author review

Abstract—Synthetic training reduces the need for task-specific real annotations, but self-training can reinforce inaccurate target-domain landmarks. We examine whether correcting pseudo-label coordinates improves monocular pallet pose estimation under matched student training conditions in two pallet material categories. The same frozen PoseFix-derived teacher, adapted using nine real images and 38 manually verified corners with synthetic replay, corrects a synthetic-pallet-pretrained model’s predictions. Within each material, raw and corrected students share images, support, initialization, augmentation settings, synthetic replay, and 320 optimizer updates; the plastic and wood pools expose 217 and 361 unique real images, respectively. On 66 directly verified visible plastic development points, teacher correction increases 10-pixel accuracy from 44 to 50 out of 66; trusted wood pseudo-coordinate quality remains unresolved. On 128 reused, recording-disjoint ordinary-plastic images, corrected versus raw student PCK10 improves from 47.51% to 51.47% and geometry-reference pose AUC from 0.33472 to 0.35902. On 45 similarly disjoint wood development images, the corresponding changes are 47.11% to 47.69% and 0.65643 to 0.66503. The wood gain is only two corners and does not surpass its original baseline. These pooled contrasts share a positive direction, but tails and recording-wise effects are mixed. Pose references are not independently measured physical ground truth. The evidence is limited to the two evaluated categories in reused development data, not universal material generalization or large-error recovery.

Index Terms—Pallet perception, monocular pose estimation, pseudo-labels, self-training, synthetic-to-real adaptation, evaluation provenance.

I. Introduction

An RGB pallet-pose estimator must localize geometric landmarks before recovering object position and orientation. Synthetic images can provide pallet-specific supervision, but real images introduce lighting, occlusion, and viewpoint shifts. A confident predicted box need not imply accurate corners. Consequently, treating every confident prediction as a training target can preserve or amplify localization errors.

The question in this work is deliberately narrower than developing an additional pose architecture: under a disclosed real-supervision budget, are corrected pseudo-label coordinates more useful than the original coordinates for student adaptation? A gain over the original synthetic-only model is insufficient to answer this question. It could instead result from additional optimizer updates, synthetic replay, or a different subset of target images. A useful comparison must hold those factors fixed and compare raw-target and corrected-target students directly.

We close the original plastic comparison and a bounded wood extension without introducing further correction modules. A frozen teacher supplies corrected coordinates, while the student is used without that teacher at inference. We distinguish label quality, corrected versus raw student value, value beyond the baseline, and evidence scope. The contributions are (i) limited-supervision pseudo-coordinate correction with a disclosed manual budget, (ii) an auditable matched raw-versus-corrected student comparison with joint 2D and fixed-pipeline 6D evaluation, and (iii) material-stratified evidence across ordinary plastic and wood, retaining recording heterogeneity, error tails, and unfavorable cases. Self-training, pose refinement, and PnP are not claimed as new algorithms.

II. Related Work and Scope

STAC uses confident teacher detections as pseudo-labels and trains a student with augmentation-based consistency[1]. This motivates separating target quality from student training. Our experiment concerns pallet corner coordinates and matched support, and does not reproduce STAC’s object-detection benchmarks or claim its annotation-efficiency results.

PoseFix refines an input human pose from the image and synthesizes training errors using human-pose error statistics[2]. Our teacher is a pallet-specific adaptation of this family, not the original human benchmark. It uses pallet landmarks, a synthetic-trained prior, limited real manual coordinates, and synthetic replay. At self-training time the teacher is frozen; there is no human correction of each pseudo-label.

Self6D demonstrates synthetic-to-real pose adaptation with differentiable rendering and unannotated real RGB-D data during adaptation[3]. Our tested correction path instead consumes RGB, a predicted box, and predicted landmarks; it does not use a depth sensor or a rendering-based adaptation objective. Registered dimensions and camera intrinsics are used only by downstream geometry and the existing consistency filter.

BOP established evaluation that explicitly addresses object symmetries and pose ambiguity[4]; subsequent challenge reports distinguish datasets, tasks, and seen versus unseen object conditions[5]. Those distinctions motivate careful reporting here. Our normalized, symmetry-aware cuboid AUC is a repository-specific geometry-reference

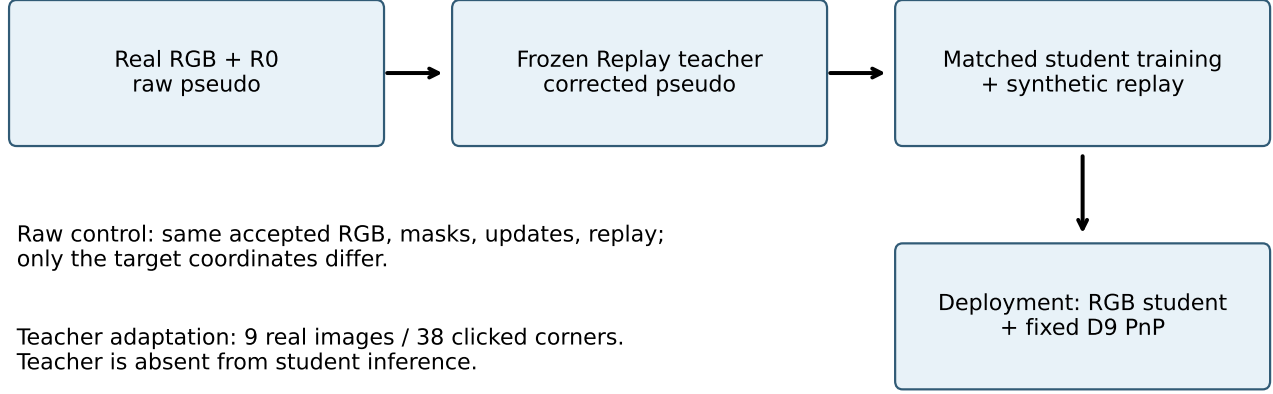


Fig. 1. The tested information paths. The teacher changes training targets, not deployed student architecture. Both students use the same accepted real images and target-support masks. The final geometry stage is fixed across arms.

measurement, not a claim of BOP benchmark equivalence or state-of-the-art performance. None of these prior papers’ numerical results is directly compared across incompatible datasets.

III. Method and Matched Intervention

A. Baseline and point contract

The baseline R0 is the existing YOLO26n pallet-pose checkpoint. “Synthetic-only” describes its pallet-specific fitting, not the provenance of all pretraining: upstream COCO-pose pretraining is documented. For each image, the fixed inference wrapper returns candidate boxes, scores, and nine landmarks. Corners 0–7 follow the registered cuboid convention; landmark 8 is the existing center. Instance selection and original-image coordinate restoration are unchanged across the comparison. Ground-truth boxes or poses do not choose an inference candidate.

B. Frozen correction teacher

The teacher is the existing RGB ResNet152 PoseFix-derived pallet network. It was initialized from the synthetic-trained PRIOR1 checkpoint and adapted for 300 updates using eight real and eight synthetic examples per update. The real term supervises only 38 verified manual corners in nine images: three ordinary-plastic night images and six wood-day images. Unknown and PnP-completed points and the center do not supply manual teacher supervision. Frozen batch-normalization statistics and trainable affine parameters follow the original protocol.

Teacher adaptation retains the existing heatmap cross-entropy plus coordinate loss and source replay objective. Real inputs alternate between R0 predictions and manually supported coordinates perturbed with predicted-box-scaled noise. We do not claim that synthetic and real errors have identical distributions. The original fixed final teacher checkpoint is reused without a new fit or development-selected displacement cap. At pseudo-label generation, it consumes only RGB, R0 points, and the

predicted crop. It preserves the detection box, score, keypoint confidence, and center, and modifies corner coordinates.

C. Shared image selection and support

The ordinary-plastic candidate pool contains 1,000 real images outside the evaluation recordings. The existing raw confidence, horizontal-flip agreement, and leave-one-out (LOO) consistency checks retain 259 candidates; applying the frozen teacher and the existing post-correction LOO check retains 249. This pipeline does not imply that an accepted prediction is correct. No new geometric-shape, X-intersection, stability, or threshold sweep is introduced.

Both training arms use this same accepted set. A fixed replacement draw produces 512 real slots containing 217 unique images per epoch. For every raw/corrected pair, the box and trusted-keypoint mask are identical. The latter is the intersection of the original target-export validity masks; points outside shared support are ignored in both arms. Thus the controlled contrast concerns corrected versus raw coordinates conditional on a shared teacher-filtered selection policy. It does not estimate the effect of teacher-based selection itself, nor does the raw arm eliminate every influence of the manually adapted teacher.

Let $\tilde{\mathbf{y}}_i^r$ and $\tilde{\mathbf{y}}_i^c$ denote the raw and corrected targets and m_{ij} their common support. With the same augmentation A , real batch B_t , and synthetic replay batch S_t , the intervention is

$$\mathcal{L}_t^a = \mathcal{L}_{\text{pose}}(f_\theta(Ax_i), A\tilde{\mathbf{y}}_i^a; m_i) + \mathcal{L}_{\text{source}}, \quad a \in \{r, c\}. \quad (1)$$

This expression describes the existing pose objective and shared support, not a newly designed loss. Identical frozen detector branches retain the same ancillary detection terms. The only altered supervised targets are the real coordinate values.

D. Student training and deployment

Each student starts independently from the same R0 checkpoint. Only the pose branches and flow parameters are trainable; backbone, neck, box/class branches, and all running buffers remain exactly fixed. Each epoch uses 512 real and 512 synthetic slots; five epochs with batch 16 give 320 updates and 2,560 exposures of each domain. The synthetic pool has 512 unique images and no negative images. Training uses AdamW, cosine decay, and the existing true-ignore keypoint implementation. The checkpoint is the fixed final checkpoint, never an epoch chosen by evaluation performance.

The historical experiment tested learning rates 10^{-4} and 10^{-5} with matched source-only, raw-target, and corrected-target arms. The latter rate was selected in the earlier reused development study. We disclose that selection, report both settings, and retain both available order repeats at the selected rate. Orders 43 and 44 change sample order; they are not three independent initialization seeds. Corresponding first-batch image-tensor hashes match within each raw/corrected repeat. This is a first-batch check, not a claim that a complete per-step tensor trace was recorded. Table XIII lists the controlled factors.

At deployment, only the student processes RGB. The teacher is absent. The final pose stage uses the same original D9 width/depth hypothesis selector for every arm and solves from corners 0–7. Landmark 8 remains part of the original selector residual; that known design choice is held fixed rather than altered to favor an arm. No trained model-specific router, reference-dependent oracle, or additional inference-time correction enters the main comparison.

IV. Evaluation Protocol

A. Populations and reference provenance

The main panel comprises 128 ordinary-plastic images from seven recording groups, with 29 Clean, 21 Moderate-occlusion, and 78 Severe-occlusion images. Its recording groups do not overlap the pseudo-label training pool or the teacher’s manual-training sessions, including recorded aliases. Nevertheless, this is repeatedly inspected development data, not independent confirmation. The historical full 194-image plastic panel included three teacher-training images; that aggregate is not substituted for the present comparison. The 300-image mixed-material panel is not pooled with these denominators.

Label-quality evaluation uses 66 directly verified visible corners in 16 images within the 128-image panel. The frozen FINAL_V2 reference retains only manually clicked, directly visible points after human metadata QA. Scoring uses fixed native point identity, without choosing a symmetry permutation. Missing points receive the native image diagonal as error. This is a selected, labeled development proxy for pseudo-label quality; it cannot establish the unknown accuracy of every accepted unlabeled adaptation image or the accuracy of hidden corners.

The full-panel legacy labels contain mixed point provenance. Their geometry-resolved poses combine image annotations, intrinsics, and registered dimensions and are not external physical 6D measurements. Because image references and pose references are related, agreement on both does not constitute two independent forms of metrological validation.

B. Metrics and fixed denominators

The main localization metric is PCK10, the fraction of all supported corners within 10 original-image pixels. PCK5 and PCK20, median error, P90, and errors above 20 pixels are also retained. These pixel thresholds are transparent operating tolerances for this dataset, not asserted universal pallet or robotics standards. Full-panel scoring uses one permitted whole-object symmetry permutation, not independent point-wise nearest matches. Fixed-identity results are retained separately. All 128 images contain detections, but only 120 meet the unchanged box-IoU matching gate of 0.5. Full-denominator PCK includes penalties for the remaining images over 985 supported corners; matched median/P90 uses 931 observed corners. Neither difficult frames nor failures are removed from PCK.

Final pose reporting includes coverage, width/depth-axis correctness, rotation and yaw error, translation error in centimeters, cuboid IoU3D, and normalized ADDsym AUC. The latter integrates the unchanged symmetry-aware error-threshold accuracy curve up to 0.1 object diameter. Failed poses count against AUC; every arm yields a pose for all 128 images here. An axis-correct hypothesis need not have accurate full rotation, and favorable AUC need not improve the median of every pose statistic.

The primary 2D contrast isolates coordinate learning; the common final pose pipeline tests its downstream value. Model-conditioned selectors and nondeployable reference oracles remain separate historical extensions. No new latency or operational-safety claim is inferred from these accuracy measurements.

C. Uncertainty and retrospective disclosure

We report paired per-frame changes, correct/incorrect corner transitions, every recording group, both learning-rate settings, and all order repeats. A 10,000-draw recording-cluster bootstrap describes PCK10 differences. Corners are not treated as independent experimental replicates. With only seven reused recording groups and historical configuration search, these intervals are exploratory and not search-adjusted. The present closure fixes the scope before new re-evaluation, but is not a prospective preregistration of the original experiments.

V. Results

A. Q1: Does the teacher improve target quality?

On the same verified plastic points, PCK10 increases from 44/66 to 50/66, and median error decreases from

TABLE I

Matched coordinate intervention within each material. Different accepted pool sizes and evaluation conditions are disclosed rather than treated as a randomized material intervention.

Item	Plastic main	Wood extension
Frozen teacher	Same Replay9images/38corners	Same checkpoint; no refit
Candidate/accepted/used unique	1000 /249 /217	1000 /676 /361
Initial student	Same R0	Same R0
Trainable state	Pose branches+flow	Same
Protected state	Backbone/detector/all buffers	Same; exact checks
RAW/REF parity	RGB/order/box/support/source	Same; coordinate values differ
Replay per epoch	512real+512synthetic slots	Same source512
Optimizer/LR/updates	AdamW /1e-5 /320	Identical
Epoch/batch/nbs/seed	5 /16 /16 /42	Identical
Selection	Fixed final last.pt	Identical
Final pose	Common D9; no teacher at deployment	Same method; Wood registered dimensions
Material routing	Externally supplied material	Externally supplied material
Evidence	Reused DEV128,7recordings	Reused DEV45,2recordings

7.144 to 5.489 pixels (Table II). The P90 and count of errors above 20 pixels also decrease. This supports improved visible-point quality on this proxy, not perfect correction or unmeasured accuracy on all training targets. No eligible directly verified wood points are available; its Q1 entry is unresolved, not zero accuracy. Figure 2 includes both an improvement and a deterioration selected from the actual paired errors.

B. Q2: Does corrected-target training improve the student?

At the previously selected learning rate, corrected-target training raises PCK10 from 47.51% to 51.47% (Table XI), a 3.96 percentage-point difference. The descriptive recording-bootstrap interval is [1.83,6.46] percentage points. The paired mean scored frame error improves on 92 images, worsens on 28, and is unchanged on eight. At the 10-pixel boundary, 51 corners become correct and 12 become incorrect. The eight unchanged frame scores include detection-gate penalty ties; they are not necessarily identical predicted coordinates. No raw corner above 20 pixels becomes correct within 10 pixels in this main student contrast. The gain should therefore not be described as successful general large-error recovery.

The pose stage also improves under identical selection and solver code: ADDsym AUC changes from 0.33472 to 0.35902 (Table XII). This is a final-pipeline result, not an oracle choice among poses. The other historical learning rate and both order repeats retain a favorable corrected-versus-raw direction for pooled PCK10 and AUC (Table VII). They support consistency within the explored settings, not independent dataset replication.

Importantly, on the smaller manually reverified subset the two students tie at 43/66 PCK10, compared with R0's 44/66 (Table IX). Corrected-student PCK20 and P90 improve there, but its median is slightly worse than the raw student's. Thus the full-panel PCK10 gain is not reproduced at that threshold on this visible-only reference. Teacher quality and student improvement are distinct

measurements; the favorable teacher result cannot replace this student check.

C. Q3: What is added beyond synthetic-only R0?

R0 obtains 49.14% PCK10 and 0.33796 pose AUC, compared with 51.47% and 0.35902 after corrected-target self-training. The raw-target student is worse than R0 on those pooled metrics, so the result is not simply a benefit of any self-training. The source-only update controls are included to expose effects of additional source optimization. No detector improvement is claimed: detector boxes, scores, and candidate order remain exactly unchanged.

There is no all-metric dominance. Matched-corner P90 is 40.902 pixels for R0, 41.958 for the raw student, and 42.085 for the corrected student. The corrected model's median rotation error and axis count do not surpass R0. AUC improves despite a worse median normalized ADDsym error, reflecting different regions of the error distribution. These are observable trade-offs, not omitted measurements.

D. Q4: Where does the result hold?

Table V retains all difficulty levels. Corrected versus raw PCK10 and AUC improve in all three strata at LR5, but corrected versus R0 Moderate PCK10 is lower. Severe pose AUC changes little relative to R0, and its median translation error remains worse. Recording-wise results are retained in Table VIII; small groups must not be interpreted as precise new-domain estimates. The strongest claim is pooled and scoped to these ordinary-plastic development recordings.

VI. Bounded Material Extension: Wood

A. Population and unchanged intervention

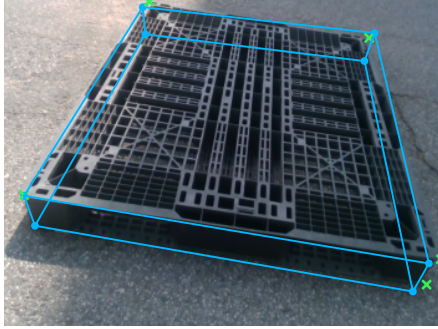
The wood extension asks whether the corrected-minus-raw contrast has the same direction beyond ordinary plastic, not whether wood performance can be improved by further tuning. Before new inference or scoring, we audited the historical 116-image wood panel. The shared

TABLE II
Trusted visible pseudo-coordinate quality. Plastic:66 points in16 reused DEV images. Wood:zero eligible independently sourced direct-visible points, so Q1 is unresolved; NA is not0%.

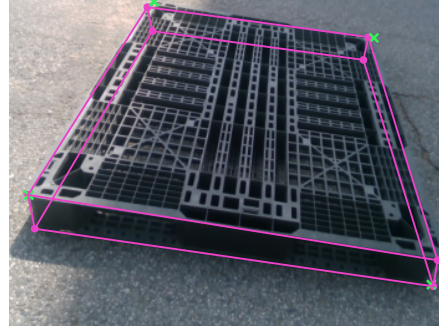
Material	Output	Trusted points	PCK5 count (%)	PCK10 count (%)	PCK20 count (%)	Med px	P90 px	>20
Plastic	Raw R0	66	21/66 (31.82)	44/66 (66.67)	60/66 (90.91)	7.144	18.390	6
Plastic	Frozen teacher	66	30/66 (45.45)	50/66 (75.76)	63/66 (95.45)	5.489	13.759	3
Wood	Raw R0	0	NA	NA	NA	NA	NA	NA
Wood	Frozen teacher	0	NA	NA	NA	NA	NA	NA

eval_cad:1778653033056483584 | verified green crosses | corrected - raw -5.46px

Synthetic-only R0 | visible mean 13.00px

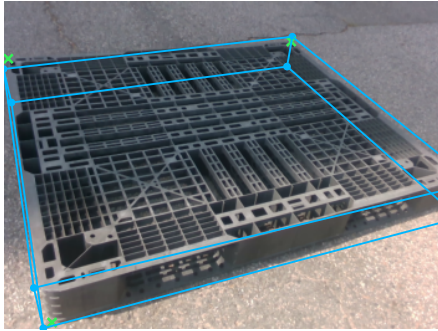


Frozen Replay teacher | visible mean 7.54px



eval_cad:1778653018878592768 | verified green crosses | corrected - raw +6.28px

Synthetic-only R0 | visible mean 13.09px



Frozen Replay teacher | visible mean 19.37px

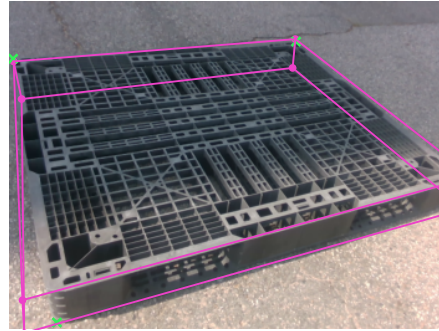


Fig. 2. Frozen raw versus corrected pseudo-label examples: one favorable and one unfavorable visible-point case. Green crosses are directly verified visible references. The actual images and predictions are reused, not synthesized; examples are selected post hoc for explanation and do not determine inference or aggregate membership.

Replay teacher used six wood-day images (23 corners) in recording REC_001 and three plastic-night images (15 corners) in REC_002. The latter recording also contains wood-night frames, despite different session names. We excluded all 71 evaluation images from these two teacher-exposed recordings, not just the five exact teacher-image overlaps. The fixed main wood panel therefore contains 45 images in REC_039 (25) and REC_042 (20): 38 Clean and seven Moderate, with no Severe group. Teacher/manual and student-pool image-ID, SHA, and recording overlap with this panel are zero. Historical reuse still makes this DEV, not independent confirmation.

Existing model-independent material review identified 13,908 wood frames. Excluding 80 images with SHA overlap with the original annotated/teacher pool left 13,828. We selected 1,000 candidates by a fixed pro-

portional day/night allocation (675/325) and midpoint-spaced sorted frame IDs, without model-output ranking. The unchanged plastic confidence, flip, LOO, teacher, and post-correction LOO pipeline retained 676. The fixed 512-slot replacement draw exposed 361 unique images. Both students share this draw, boxes, intersected support, and the original 512-image synthetic replay. Each starts from R0 and uses the original pose/flow-only AdamW, learning rate 10^{-5} , seed 42, five epochs, and exactly 320 updates. Only real pseudo-coordinate values differ. The original source-only LR5 control is material-independent and is reused without a third fit.

The same Replay9/38 checkpoint is used for both material pools. Wood is not assigned a type-specific replacement teacher. Every saved student retains the protected backbone, detector, and running buffers exactly.

TABLE III

Matched material-stratified self-training. PCK uses the full supported-corner denominator; median is conditional on matched detection. Plastic retains the original320-update main comparison, not the later640-update test.

Material	Method	Images	Corners	PCK10 %	Med px	ADDsym AUC
Plastic	R0	128	985	49.14	9.565	0.33796
Plastic	Raw ST	128	985	47.51	9.961	0.33472
Plastic	Corrected ST	128	985	51.47	8.954	0.35902
Wood	R0	45	346	48.27	10.453	0.67050
Wood	Raw ST	45	346	47.11	10.639	0.65643
Wood	Corrected ST	45	346	47.69	10.623	0.66503

TABLE IV

Common deployable D9 final pose. References are geometry-derived legacy annotations, not independent physical6D measurements. Axis correctness tests W/D parity, not complete rotation correctness.

Material	Method	Pose coverage	Axis correct	R med/P90 deg	Yaw med/P90 deg	t med/P90 cm	IoU3D med	ADDsym AUC
Plastic	R0	128/128	83/128	3.530 / 88.733	2.756 / 88.533	9.353 / 79.122	0.5865	0.33796
Plastic	Raw ST	128/128	81/128	3.942 / 89.052	2.626 / 88.600	9.226 / 77.884	0.5808	0.33472
Plastic	Corrected ST	128/128	82/128	3.766 / 88.809	2.550 / 88.656	9.186 / 78.938	0.5928	0.35902
Plastic	Source-only update	128/128	83/128	3.507 / 88.681	2.590 / 88.341	9.203 / 78.605	0.5948	0.33895
Wood	R0	45/45	40/45	1.545 / 9.190	0.471 / 3.583	2.095 / 8.011	0.7898	0.67050
Wood	Raw ST	45/45	40/45	1.598 / 9.084	0.536 / 3.469	2.349 / 8.259	0.7704	0.65643
Wood	Corrected ST	45/45	40/45	1.605 / 9.337	0.499 / 3.588	2.071 / 7.985	0.7740	0.66503
Wood	Source-only update	45/45	40/45	1.612 / 9.277	0.613 / 3.271	2.195 / 7.963	0.7713	0.65812

TABLE V

All severity strata retained. Improved pooled accuracy does not imply improved tails or all pose statistics.

Group	N	Arm	PCK10 %	P90 px	ADDsym AUC	t cm
CLEAN	29	R0	65.94	17.485	0.71474	2.930
CLEAN	29	RAW_LR5	65.50	16.772	0.73053	2.818
CLEAN	29	REF_LR5	69.00	17.150	0.78126	2.218
MODERATE	21	R0	58.44	24.119	0.47955	5.907
MODERATE	21	RAW_LR5	55.19	24.087	0.47131	6.376
MODERATE	21	REF_LR5	56.49	23.008	0.51188	6.242
SEVERE	78	R0	40.37	53.725	0.15976	13.677
SEVERE	78	RAW_LR5	38.70	55.113	0.15079	14.692
SEVERE	78	REF_LR5	43.52	55.959	0.16087	14.431

No additional labels, filter tuning, new loss, or severity-specific routing is introduced. The deployment model is material-specific; material type is externally provided for routed evaluation. Both materials use the same deployable D9 selection principle and registered material dimensions. All five wood-arm native predictions and D9 poses were hash-locked before scoring references were opened by the scorer.

The wood panel has 346 supported legacy corners. All 45 images are detected and matched, and all arms return poses for 45/45. Its corner provenance does not establish any eligible independently sourced directly visible clicks. Consequently, trusted wood pseudo-label quality is unresolved; legacy teacher scores do not fill that gap. Its 6D

reference is geometry-derived rather than independently measured physical pose.

B. Results and scope

Tables III and IV retain within-material baselines and matched students. Wood corrected versus raw PCK10 changes from 163/346 to 165/346 (47.11% to 47.69%; +0.578 percentage points), with eight correct entries and six exits at 10 pixels. No corner above 20 pixels is recovered within 10 pixels. Mean scored frame error improves on 22 images and worsens on 23. Median corner error changes from 10.639 to 10.623 pixels, whereas P90 worsens from 67.839 to 68.985 pixels and PCK20 stays at 75.43%. These

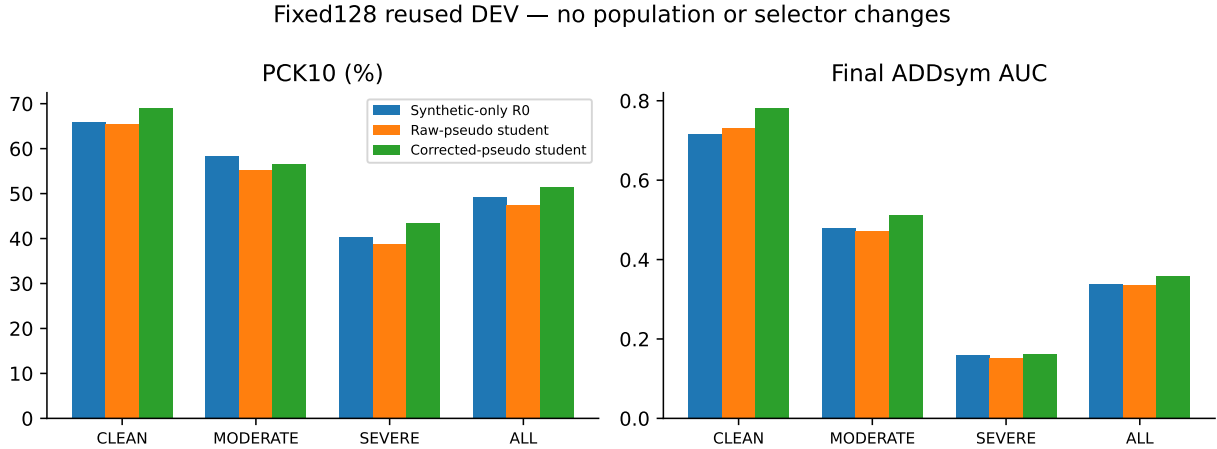


Fig. 3. All pre-existing severity strata, under the same evaluation contract. Pooled gains do not establish uniform tail-error recovery or independent generalization.

are small, mixed changes rather than broad corner-error recovery.

Wood common-D9 AUC changes from 0.65643 to 0.66503 (+0.00860). Rotation median/P90 changes from 1.598/9.084 to 1.605/9.337 degrees, while translation median/P90 improves from 2.349/8.259 to 2.071/7.985 cm. Axis correctness remains 40/45. The corrected student exceeds the source-only update control (47.40%, 0.65812), but not R0 (167/346, 48.27%, 0.67050). Thus the extra value over raw targets should not be conflated with a demonstrated improvement over synthetic-pallet-only R0 in wood.

Wood Clean PCK10 is tied at 143/301; Moderate increases from 20/45 to 22/45. AUC improves in both available strata. Recording-wise PCK10 increases from 48.13% to 50.80% in REC_039 but decreases from 45.91% to 44.03% in REC_042; AUC increases in both. Leaving out either recording reduces to the other one, so the sign sensitivity is explicit. Two reused recording groups do not support a reliable new-domain confidence claim, and corners are not treated as independent replicates.

The prespecified pooled-direction decision is therefore material-general signal, limited to the two evaluated categories. It is not evidence of material invariance, consistent gains in every recording, or verified wood pseudo-label quality. The historical material-routed experiments with type-specific teachers and Clean19 S0/S1/S2 used different supervision and interventions and remain supplementary context, not matched causal controls. The original plastic tables, LR/order sensitivity, verified66 tie, and bounded640-update study are preserved. No third option or further estimator experiment is introduced to enlarge this result.

VII. Limitations and Evidence Closure

First, the teacher uses real manual supervision. Our intervention is conditional on nine adaptation images and 38 corners, as well as a shared teacher-filtered training selection. We do not claim zero real labels, superiority to

directly spending the same annotation budget on the student, or a measured reduction in human annotation time. The 66 verified evaluation points are additional human effort and never student-training targets. Generic pretraining and earlier development annotation/inspection also remain part of the full project history.

Second, the real adaptation data and evaluation recordings are disjoint, but the evaluation panel has been repeatedly used for method development. The LR5 setting was chosen in that history. Session-disjoint fitting does not undo that repeated inspection. The reserve provenance audit found no currently certified untouched, labeled confirmation panel; filename labels such as FINAL are not proof of independence. We therefore close a development-only manuscript without reporting an independent-confirmation number. A future untouched confirmation should be evaluated once with the locked pipeline, without using it to select the learning rate, filters, or checkpoint.

Third, the selected visible-point reference and mixed-provenance full-panel reference answer different questions. Neither proves hidden-corner accuracy or independent physical pose accuracy. The matched comparison covers only ordinary rectangular plastic and one wood category, not square green pallets or unseen materials. Wood has only two evaluation recordings, no Severe stratum, and no eligible independently sourced visible-point quality reference. Its small pooled corrected-minus-raw gain co-exists with a worse keypoint P90 and failure to exceed R0. Material type is externally provided for routed evaluation; automatic material recognition and unknown-material handling are not evaluated. Related Clean19 occlusion-transfer, manually supervised hard-image, and model-specific selector studies involve different budgets or interventions and are historical extensions, not added to this causal contrast.

Finally, the pose-head-only matched experiment does not show that arbitrary whole-model fine-tuning or other teacher architectures will improve. P90 and severe errors

TABLE VI

Within-material corrected-minus-raw contrasts. Absolute differences between material rows do not isolate a causal material-difficulty effect.

Material	Correct10 delta	PCK10 delta pp	AUC delta	Med delta px	P90 delta px
Plastic	39	+3.959	+0.02429	-1.008	+0.128
Wood	2	+0.578	+0.00860	-0.017	+1.145

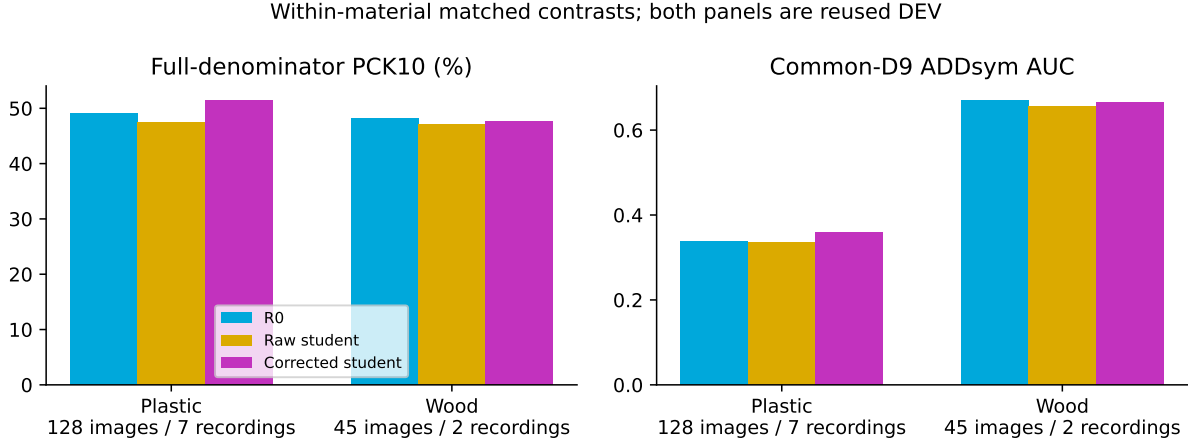


Fig. 4. Material-stratified pooled results. Corrected-minus-raw PCK10 and common-D9 AUC are positive in both panels, but the wood effect is small and below R0. Absolute plastic/wood score differences combine different images, references, viewpoint distributions, and severity coverage; they do not isolate a causal material-difficulty effect.

remain unresolved. Figure 5 explicitly includes a worsened case and a low-change case. No additional architecture, loss, selector sweep, or annotation round is triggered to remove these limitations. The appropriate next action is manuscript review and, if required, independent confirmation, rather than open-ended method search.

VIII. Conclusion

With identical target images, support, initialization, augmentation settings, synthetic replay, and update budget within each material, corrected pseudo-label coordinates yield positive pooled PCK10 and fixed-pipeline pose-AUC differences over raw-target self-training in the two evaluated material categories. The plastic student also improves over its original baseline; the wood student does not. Under the same nine-image, 38-corner teacher budget, this is a limited cross-category development signal, not evidence of universal material generalization, uniform tail recovery, or independently measured physical-pose accuracy. Wood pseudo-coordinate quality remains unresolved on trusted visible references. We close the bounded experiment with these limitations, without additional teacher, threshold, or architecture searches.

Appendix A

Sensitivity, Failure Cases, and Reproduction

Table VII contains all twelve existing student fits; no new fitting was performed in the original evidence-closure pass. A subsequently bounded, training-motivated budget experiment is reported separately below and

TABLE VII

Complete historical grid and order sensitivity. SYN is additional source-only training. ORDER43/44 change data order, not initialization seed. No best-repeat selection.

Arm	PCK10 %	P90 px	ADDsym AUC
R0	49.14	40.902	0.33796
SYN_LR4	47.41	42.356	0.33443
RAW_LR4	47.31	42.343	0.34073
REF_LR4	50.76	42.569	0.35311
SYN_LR5	48.63	41.776	0.33895
RAW_LR5	47.51	41.958	0.33472
REF_LR5	51.47	42.085	0.35902
SYN_ORDER43	48.93	41.762	0.33945
RAW_ORDER43	48.32	41.808	0.33799
REF_ORDER43	51.78	42.653	0.36100
SYN_ORDER44	48.83	41.760	0.33994
RAW_ORDER44	47.72	41.951	0.33684
REF_ORDER44	51.27	42.294	0.35880

does not replace that main comparison. Table VIII retains every evaluation recording. The repository package provides checkpoint and source hashes, full optimizer and augmentation arguments, split and target-support contracts, common-selector predictions, numerical provenance for every table, and the paired audit. Exact private RGB/annotation access is needed for full regeneration; hash manifests alone are not a public dataset release. Code and report location: <https://github.com/Canele452/pallet-6d-pose>, namespace `pallet_selftraining_paper_closure_v1`.

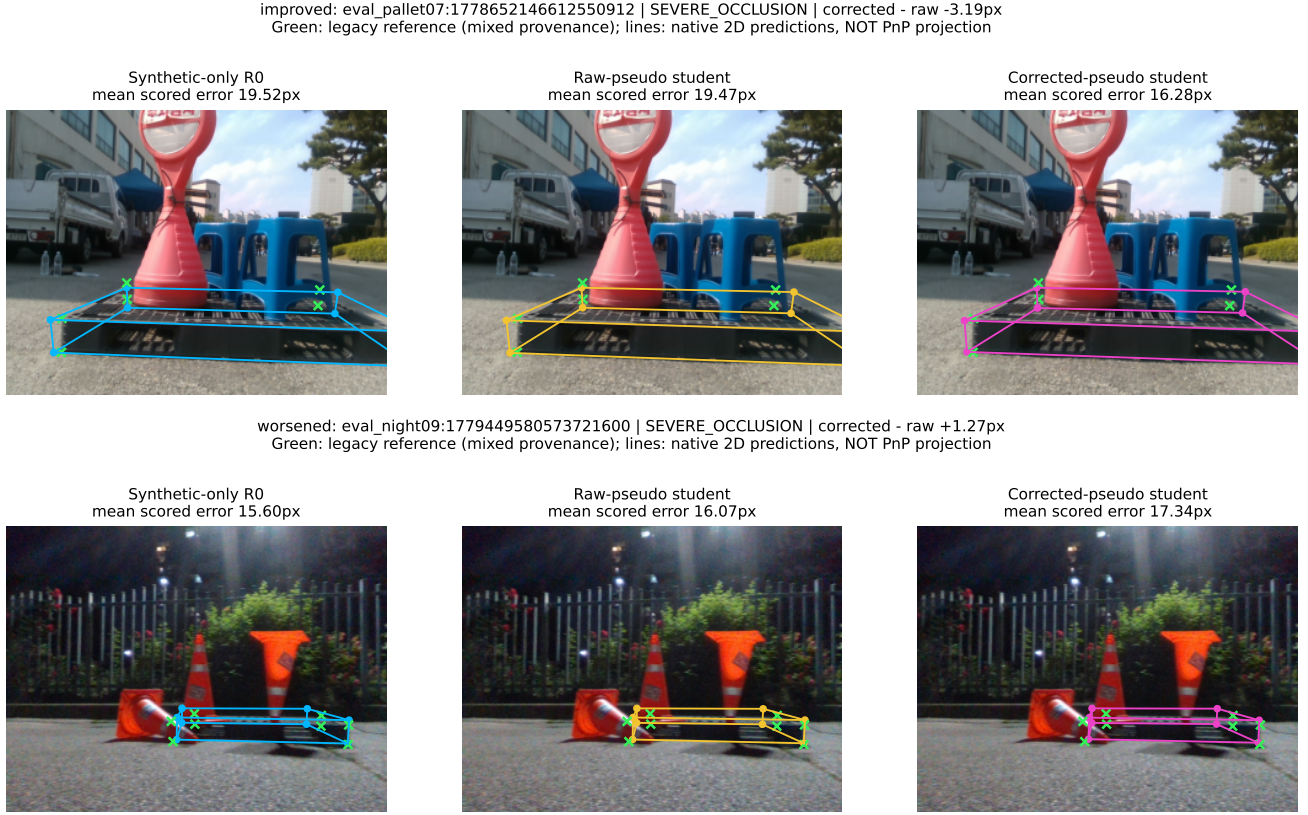


Fig. 5. Actual student examples with strong improvement (top) and deterioration (bottom) in mean scored corner error. The cyan/yellow/magenta native prediction overlays are R0/raw/corrected students, respectively; green crosses are legacy references of mixed provenance. These are raw 2D predictions, not PnP reprojections. Full-panel error scoring uses the frozen whole-object symmetry contract; image lines retain native identity.

TABLE VIII

Every reused recording group, including small or adverse groups.
Percentages are descriptive, not independent frame trials.

Recording	N	Raw PCK10	Corr PCK10	Raw AUC	Corr AUC
REC_007	33	47.71	49.24	0.08383	0.09958
REC_021	18	39.71	45.59	0.80000	0.81567
REC_022	16	32.79	36.89	0.22866	0.24034
REC_025	27	38.07	44.16	0.36820	0.38743
REC_027	12	26.09	36.96	0.14642	0.12592
REC_041	10	71.25	71.25	0.23680	0.31110
REC_044	12	96.88	96.88	0.66275	0.75483

TABLE IX

Verified-visible student sensitivity: both students tie at PCK10; the legacy-panel gain is not reproduced at this threshold on the small manually reverified subset. Same66 points, fixed identity.

Arm	PCK10	PCK20	Median px	P90 px
R0	44/66	60/66	7.144	18.390
RAW_LR5	43/66	60/66	7.046	19.574
REF_LR5	43/66	63/66	7.097	17.343

Appendix B

Post-Hoc Visible-Transfer Diagnosis and Bounded Budget Test

This analysis retains the original R0, raw LR5, corrected LR5, and nine-image/38-corner Replay teacher. No new

RGB, manual coordinates, teacher fit, selector, reliability filter, or refiner architecture was introduced. The 66 points were already inspected development data; neither freezing predictions again nor their absence from student training makes them an independent test.

A. Why the verified-point result tied

Across all 66 fixed-identity points, the original raw-to-corrected student contrast has three entries into and three exits from the 10-pixel correct set. The net changes are +2 Clean, -2 Moderate, and 0 Severe. Continuous error improves at 34 points and worsens at 32, with a mean difference of -0.461 pixels. Equal PCK10 therefore does not establish equivalent predictions. Each student has ten points where only the teacher is correct and three where only the student is correct; teacher correctness at an evaluation point is not evidence that the point was a training target.

Using identical predictions, point IDs, native coordinates, and missing-value handling, the same 66 points score 45/66 raw versus 44/66 corrected under the legacy reference, and 43/66 versus 43/66 after verification. The median legacy-verified reference displacement is 2.828 pixels (P90 5.374). Thus the corrected student's full-panel advantage was already absent on this selected subset

TABLE X

Pseudo-label quality on 66 verified visible points in 16 reused DEV images; fixed native identity. A labeled proxy, not accuracy measured on unlabeled adaptation images.

Output	Points	PCK5 %	PCK10 %	PCK20 %	Med px	P90 px	Above20
Synthetic-only R0	66	21/66 (31.82)	44/66 (66.67)	60/66 (90.91)	7.144	18.390	6
Frozen Replay teacher	66	30/66 (45.45)	50/66 (75.76)	63/66 (95.45)	5.489	13.759	3

least_changed: eval_night08:1779449483432542720 | SEVERE_OCCLUSION | corrected - raw +0.00px
 Green: legacy reference (mixed provenance); lines: native 2D predictions, NOT PnP projection

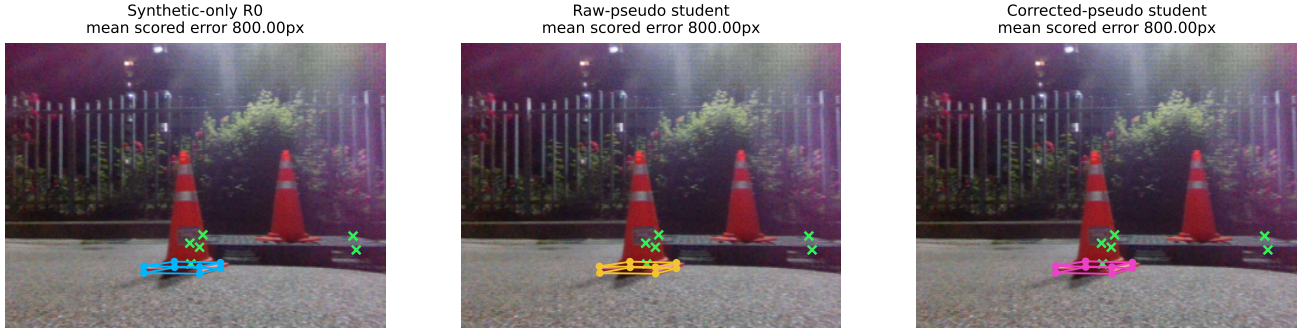


Fig. 6. A low-change scored case retained alongside favorable and unfavorable examples. Identical detection penalties can yield an unchanged score even if point coordinates move; no inference correction was chosen using these references. The Markdown report contains all 13 paired example images.

TABLE XI

Main 2D comparison on 128 images and 985 supported corners. PCK includes detection failures; medians and P90 are conditional on matched detection. LR5 was historically DEV-selected.

Arm	PCK5 %	PCK10 %	PCK20 %	Med px	P90 px
R0	20.81	49.14	72.69	9.565	40.902
RAW_LR5	20.30	47.51	72.59	9.961	41.958
REF_LR5	24.67	51.47	73.91	8.954	42.085

before reference replacement. The full-panel/subset difference cannot be attributed entirely to annotation error. No reference was automatically repaired, and PCK10 remains the principal visible-point threshold.

B. Training-target following and locked intervention

The actual real-image occurrence list contains 217 unique images and 512 slots per epoch. Among 1,627 commonly supported corners, mean residual to the corrected target is 4.078 pixels for the raw student and 3.070 for the corrected student. For the 1,492 corners with target displacement at least one pixel, 82.64% of student differences have positive projection onto the target-correction direction; the median scalar projection fraction is 0.319. This describes partial target following, not physical accuracy: the unlabeled training targets are not ground truth. Unique-image-uniform and occurrence-weighted summaries, center-only results, and recording/corner/support strata are retained separately in the repository.

The diagnostic uses unaugmented native images and highest-confidence instances, never target-nearest candi-

date replacement. Saved padded training RGB is bit-exact with the original single 100-pixel reflection padding, and the target inverse transform agrees within 10^{-5} pixels. Recomputed R0 outputs agree with raw targets to below 10^{-6} pixels on average. There is no exhaustive augmented-tensor cache. Training losses mix source and real examples, use different area/support reductions, and are not native pixel residuals. The original coordinate loss decreases overall and in its final interval but is not monotonic. Remaining target residual and late loss decrease motivated one budget-only hypothesis test; contradictory pseudo supervision, augmentation effects, replay, and frozen representation remain competing explanations.

Both new students start from the identical R0 and run 640 optimizer updates, preserving the original first 320-update learning-rate and end-to-end loss-weight schedules. The following 320 updates hold the actual final-prefix learning rate, $1.859423525 \times 10^{-6}$. Simply changing the cosine horizon to ten epochs would also change the prefix and was not used. For each arm, first-prefix training-loss/LR CSV values and saved EMA weights reproduce the original run exactly; all 747 protected tensors remain exact. This is not a claim of an exhaustive historical training-tensor audit. All images, boxes, common support, order, augmentation settings, and optimizer parameters remain matched, with only raw/corrected target coordinates differing. Both fixed final checkpoints are used. Their predictions and unchanged D9 poses were hash-locked before reference scoring.

TABLE XII

Final 6D under the SAME frozen D9 selector and corner-only solver. Reference is geometry-derived, not independently measured physical pose.

Arm	Pose	Axis	R deg	Yaw deg	t cm	IoU3D	ADDsym AUC
R0	128/128	83/128	3.530	2.756	9.353	0.5865	0.33796
RAW_LR5	128/128	81/128	3.942	2.626	9.226	0.5808	0.33472
REF_LR5	128/128	82/128	3.766	2.550	9.186	0.5928	0.35902

TABLE XIII

Matched coordinate intervention. Shared image membership is conditioned on the same raw-plus-refined filter acceptance, so this is not a comparison of different selection policies.

Contract	RAW and corrected
Initialization	Same R0 checkpoint
Trainable state	Pose branches and flow only
Real RGB	217 identical unique images
Real/synthetic exposure	2560 / 2560 each
Synthetic pool	512 identical images; no negatives
Supervised support	Identical raw/ref confidence intersection
Optimizer / updates	AdamW / 320
Learning rate	1e-5 main; 1e-4 sensitivity
Augmentation / seed	Same settings / 42
Checkpoint choice	Fixed final last.pt
Teacher supervision	9 images / 38 manual corners
Only changed target	Supervised pseudo coordinates

C. Mixed result and stopping decision

Both students gain one verified correct point, so the corrected-versus-raw tie remains (Table XVII). Corrected 640 versus corrected 320 has one entry and no exit at PCK10, while corrected versus raw at 640 has two entries and two exits. Corrected PCK5/PCK20 remains 26/66 and 63/66. Its mean verified error decreases from 10.082 to 10.040 pixels and P90 from 17.343 to 17.005, but median error worsens from 7.097 to 7.120. The corrected training-target residual falls by only 0.076 pixels. These small changes do not establish that additional optimization resolved visible transfer.

On the unchanged full panel, corrected 640 versus corrected 320 loses three net correct corners (507/985 to 504/985; two entries, five exits), while pose AUC rises from 0.35902 to 0.36189. Matched P90 worsens from 42.085 to 42.160 pixels. Under the same 640-update budget, corrected still exceeds raw by 3.553 PCK10 percentage points and 0.02509 AUC, compared with 3.959 points and 0.02429 at 320 updates. Thus the intervention’s effect and the corrected-target added value are different contrasts: the former is mixed, and the latter remains favorable on this reused legacy panel but not superior at verified PCK10. Detection remains 128/128, matched detection 120/128, and pose coverage 128/128 for all arms. The original 32-image synthetic validation role and all severity/recording summaries are retained without calibration or checkpoint selection.

This single paired experiment does not uniquely identify the remaining cause or rule out every possible

training budget. It provides no independent-session, hidden-point, or physical-6D validation. We preserve the original main results and close method development without a second intervention, additional labels, or a new uncertainty module. Full contrasts, damaging and low-change examples, hashes, and reproduction instructions are provided in the repository namespace `pallet_visible_transfer_closure_v1`.

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TABLE XIV

Full2D results and the reused source-only update control. Med/P90 are matched-only; gross errors and PCK retain the full supported denominator.

Material	Method	Detected	Matched	Correct10	PCK5 %	PCK10 %	PCK20 %	Med px	P90 px	>20
Plastic	R0	128/128	120/128	484/985	20.81	49.14	72.69	9.565	40.902	269
Plastic	Raw ST	128/128	120/128	468/985	20.30	47.51	72.59	9.961	41.958	270
Plastic	Corrected ST	128/128	120/128	507/985	24.67	51.47	73.91	8.954	42.085	257
Plastic	Source-only update	128/128	120/128	479/985	20.61	48.63	72.79	9.741	41.776	268
Wood	R0	45/45	45/45	167/346	17.34	48.27	76.01	10.453	66.777	83
Wood	Raw ST	45/45	45/45	163/346	16.47	47.11	75.43	10.639	67.839	85
Wood	Corrected ST	45/45	45/45	165/346	17.34	47.69	75.43	10.623	68.985	85
Wood	Source-only update	45/45	45/45	164/346	17.34	47.40	74.86	10.570	68.191	87

TABLE XV

All available severity strata. Wood has Clean38 and Moderate7; Wood severe is unavailable and is not represented as zero.

Material	Severity	N	Method	Correct10	PCK10 %	P90 px	ADDsym AUC
Plastic	CLEAN	29	R0	151/229	65.94	17.485	0.71474
Plastic	CLEAN	29	Raw ST	150/229	65.50	16.772	0.73053
Plastic	CLEAN	29	Corrected ST	158/229	69.00	17.150	0.78126
Plastic	MODERATE	21	R0	90/154	58.44	24.119	0.47955
Plastic	MODERATE	21	Raw ST	85/154	55.19	24.087	0.47131
Plastic	MODERATE	21	Corrected ST	87/154	56.49	23.008	0.51188
Plastic	SEVERE	78	R0	243/602	40.37	53.725	0.15976
Plastic	SEVERE	78	Raw ST	233/602	38.70	55.113	0.15079
Plastic	SEVERE	78	Corrected ST	262/602	43.52	55.959	0.16087
Wood	CLEAN	38	R0	146/301	48.50	64.743	0.66303
Wood	CLEAN	38	Raw ST	143/301	47.51	65.697	0.64861
Wood	CLEAN	38	Corrected ST	143/301	47.51	67.236	0.65745
Wood	MODERATE	7	R0	21/45	46.67	337.336	0.71107
Wood	MODERATE	7	Raw ST	20/45	44.44	336.781	0.69893
Wood	MODERATE	7	Corrected ST	22/45	48.89	334.899	0.70621

TABLE XVI

All recording/session contrasts, including unfavorable groups. Session and recording summaries are overlapping descriptions, not additional independent samples.

Material	Recording/session	N	Raw PCK10 %	Corr PCK10 %	Raw AUC	Corr AUC
Plastic	REC_007	33	47.71	49.24	0.08383	0.09958
Plastic	REC_021	18	39.71	45.59	0.80000	0.81567
Plastic	REC_022	16	32.79	36.89	0.22866	0.24034
Plastic	REC_025	27	38.07	44.16	0.36820	0.38743
Plastic	REC_027	12	26.09	36.96	0.14642	0.12592
Plastic	REC_041	10	71.25	71.25	0.23680	0.31110
Plastic	REC_044	12	96.88	96.88	0.66275	0.75483
Wood	REC_039	25	48.13	50.80	0.77990	0.78514
Wood	REC_042	20	45.91	44.03	0.50210	0.51490
Wood	SESSION_wood_183705	25	48.13	50.80	0.77990	0.78514
Wood	SESSION_wood_184309	20	45.91	44.03	0.50210	0.51490

TABLE XVII

Bounded post-hoc budget test, not a replacement for the 320-update main experiment. TRAIN residual is to the corrected pseudo target, not to real ground truth. Full-panel PCK10 retains all 985 supported corners.

Arm / updates	PCK10 /66	Full (%)	AUC	Fit px
Raw / 320	43/66	47.513	0.33472	4.078
Corrected / 320	43/66	51.472	0.35902	3.070
Raw / 640	44/66	47.614	0.33681	4.032
Corrected / 640	44/66	51.168	0.36189	2.994